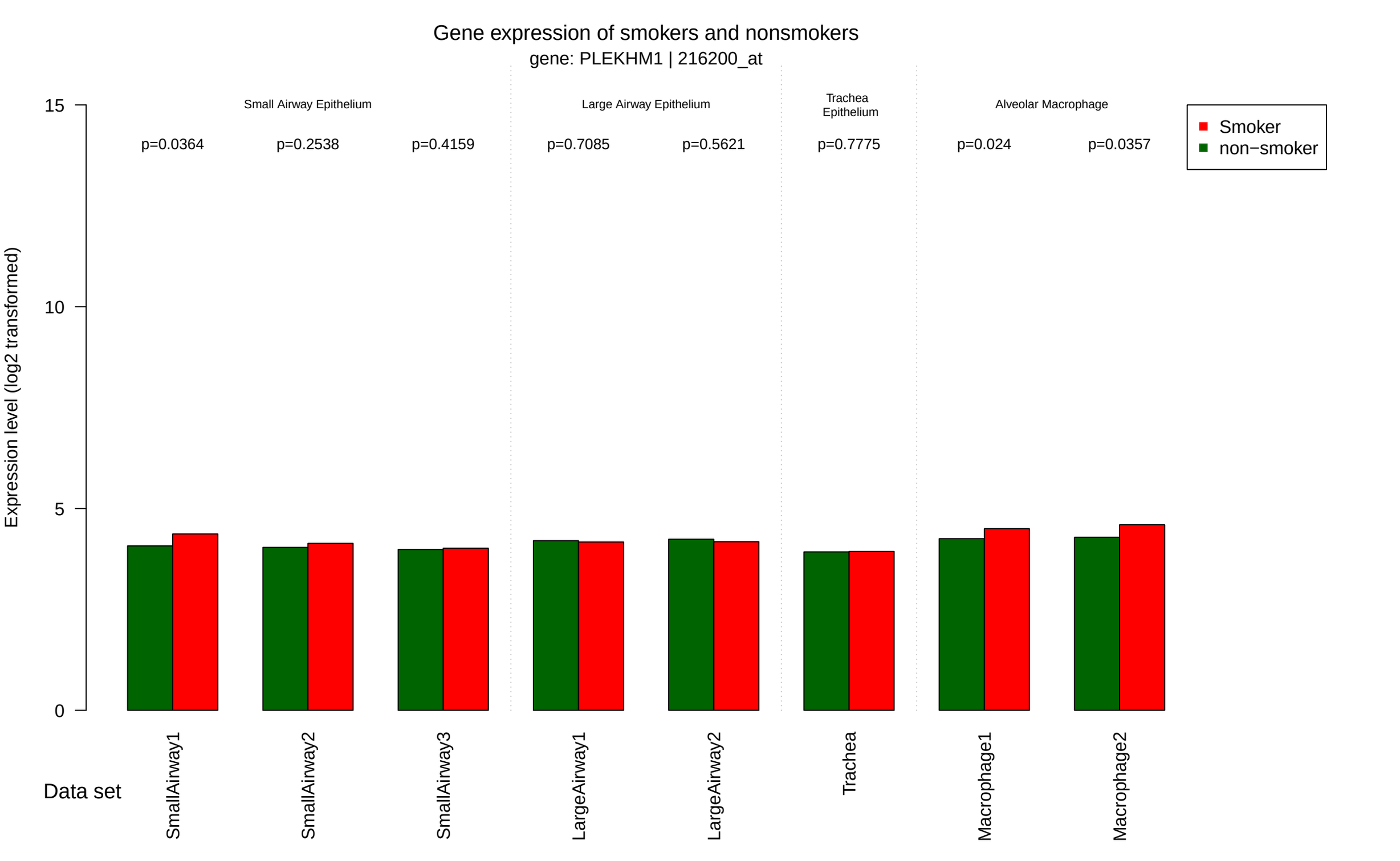
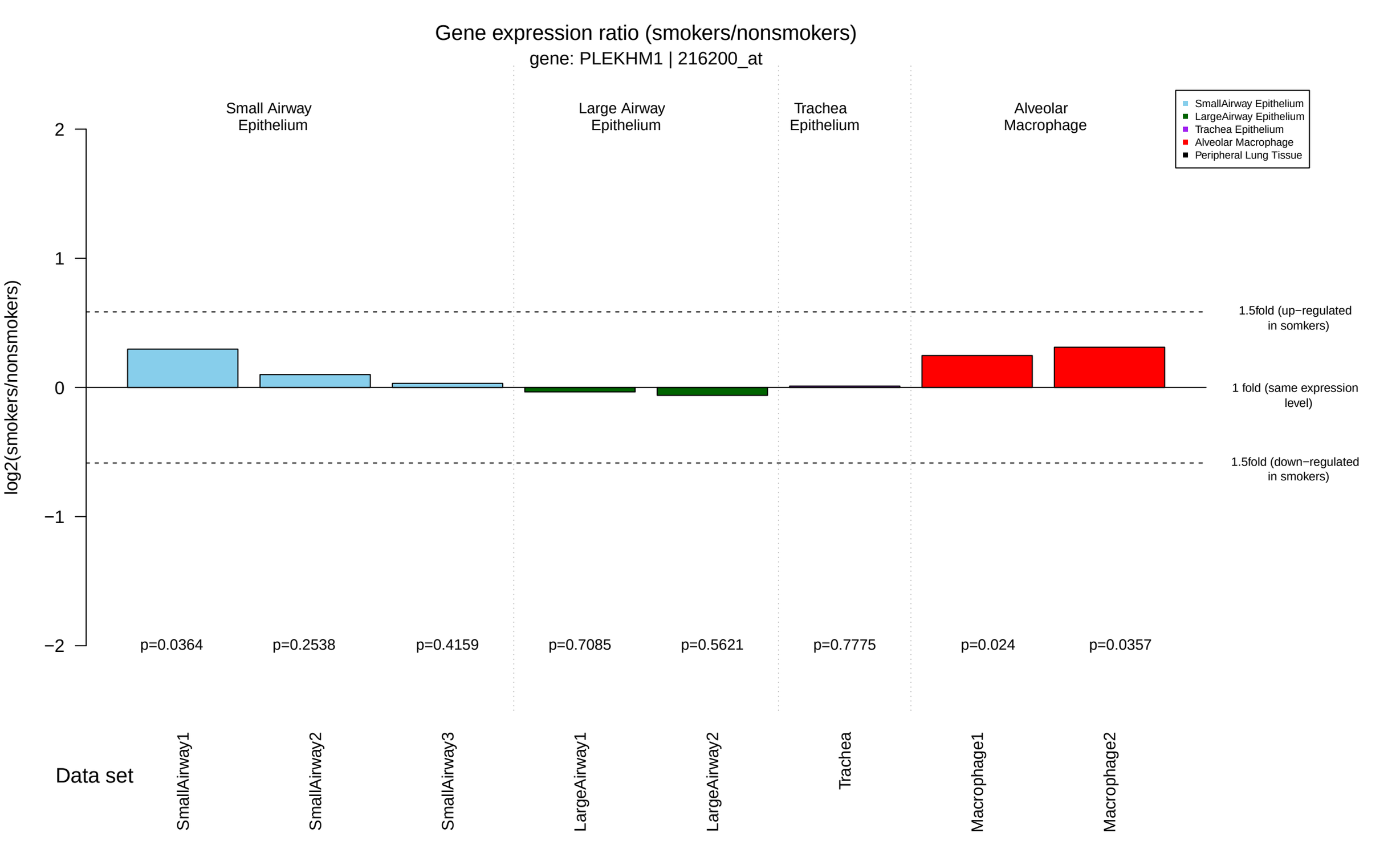
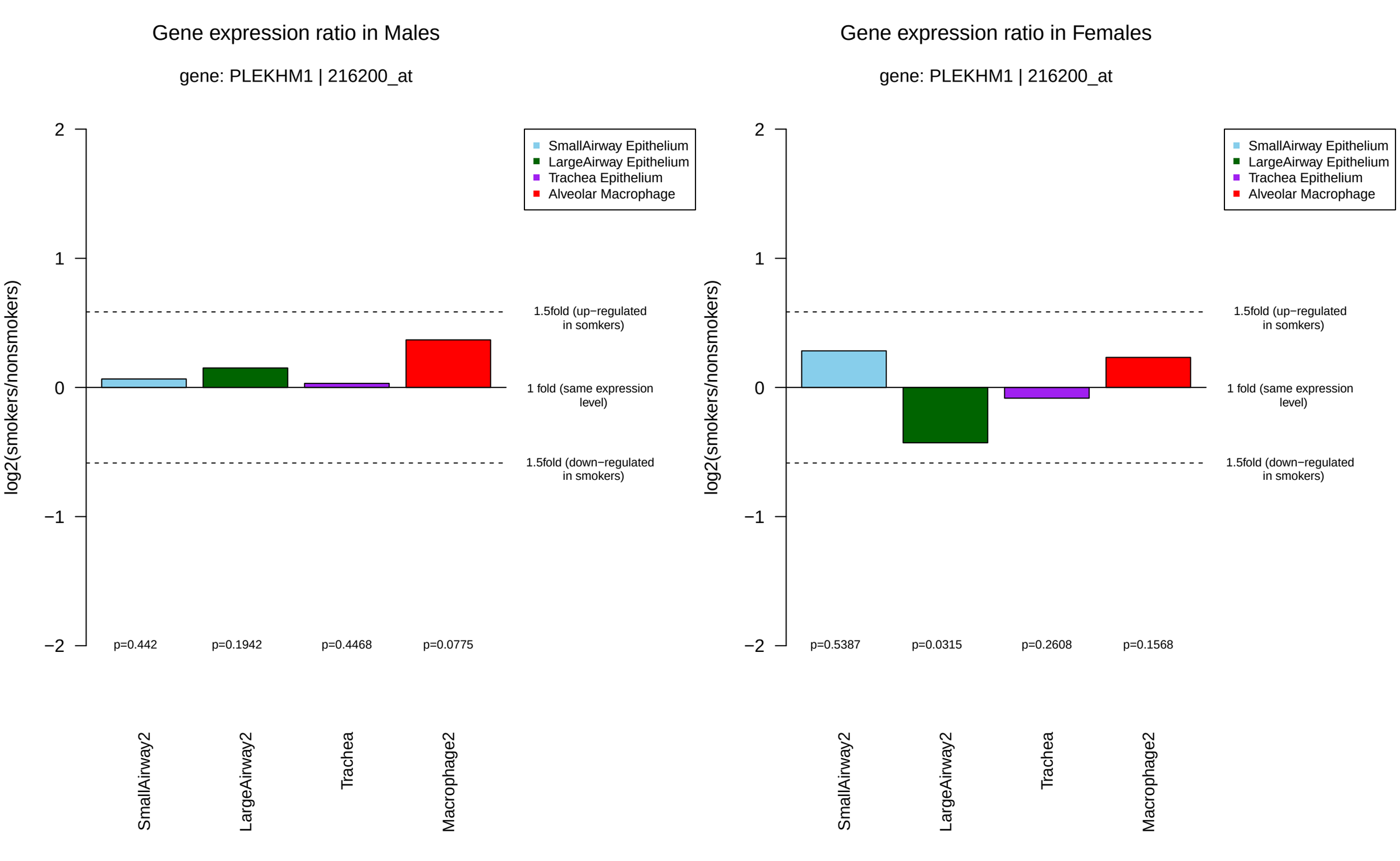
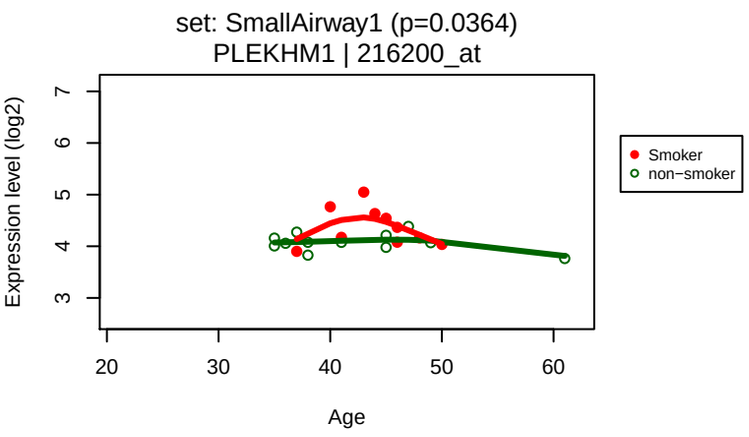
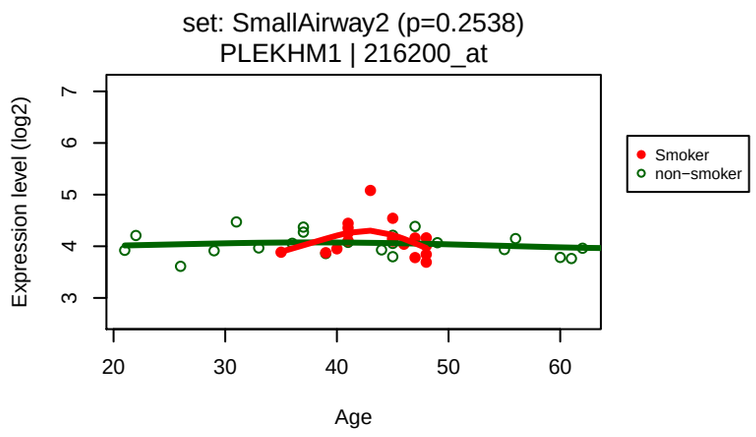
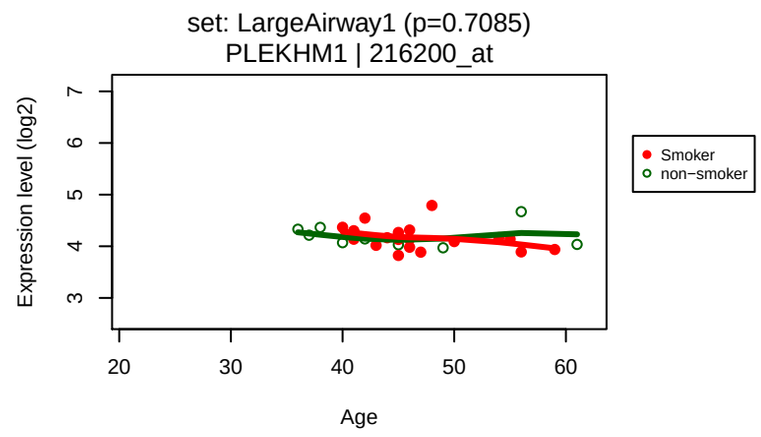

Gene expression of smokers and nonsmokers
gene: PLEKHM1 | 216200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0364
p=0.2538
p=0.4159
p=0.7085
p=0.5621
p=0.7775
p=0.024
p=0.0357
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

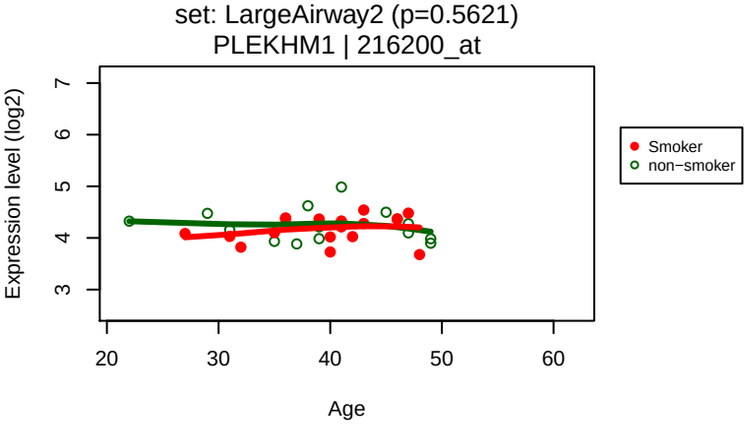

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHM1 | 216200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0364
p=0.2538
p=0.4159
p=0.7085
p=0.5621
p=0.7775
p=0.024
p=0.0357
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

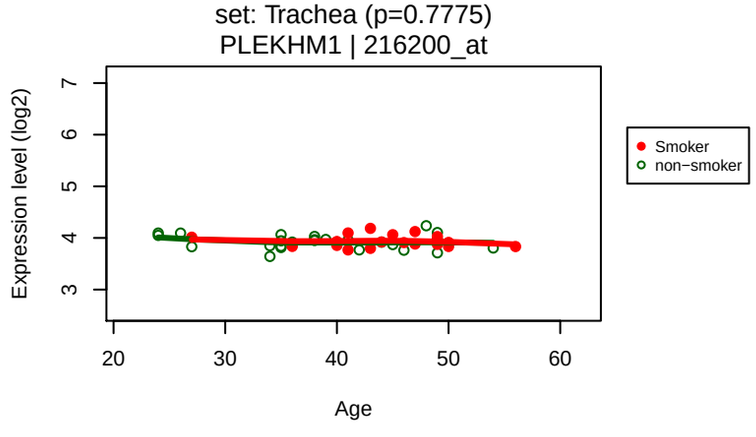

Gene expression ratio in Males
gene: PLEKHM1 | 216200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.442
p=0.1942
p=0.4468
p=0.0775
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHM1 | 216200_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5387
p=0.0315
p=0.2608
p=0.1568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

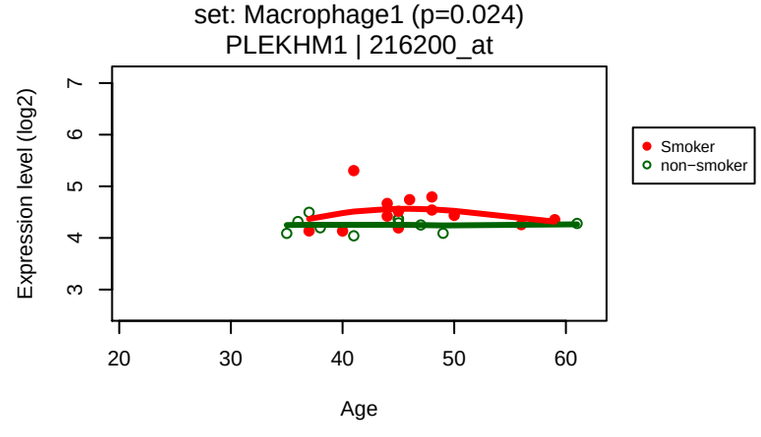

set: SmallAirway1 (p=0.0364)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker

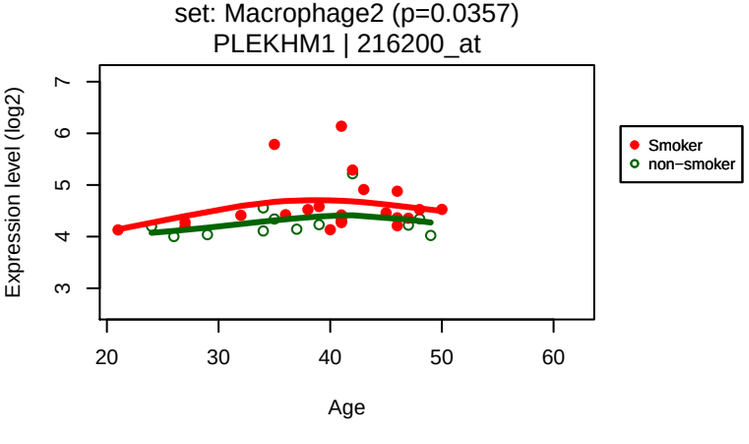

set: SmallAirway2 (p=0.2538)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker

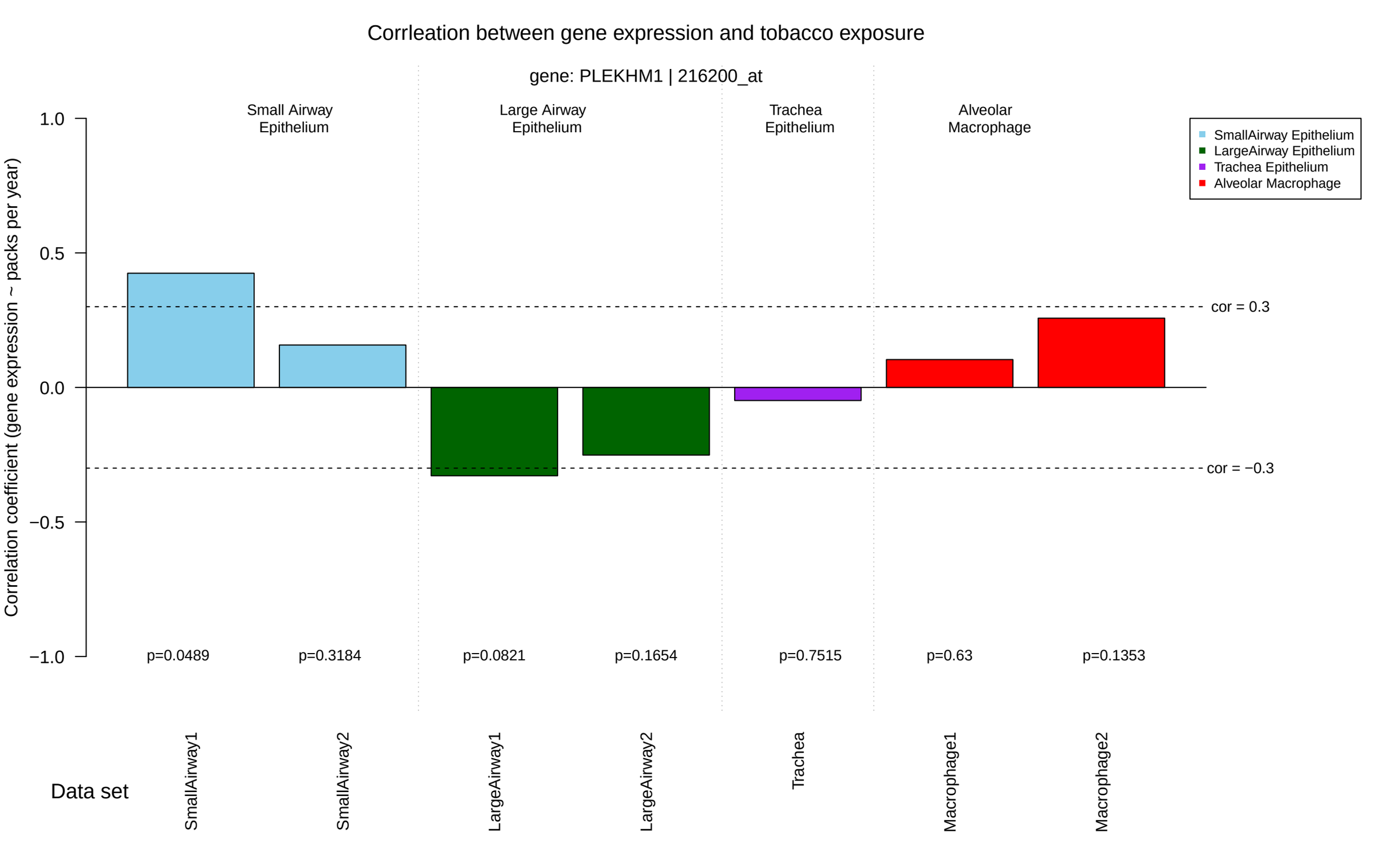

set: LargeAirway1 (p=0.7085)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5621)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7775)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.024)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0357)
PLEKHM1 | 216200_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHM1 | 216200_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0489
p=0.3184
p=0.0821
p=0.1654
p=0.7515
p=0.63
p=0.1353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2